

1.6 Sequences & Series

Mathematics B (MEI) — OCR B — A-Level (H640) — Pure Mathematics — spec refs s1, binomial expansion

What this topic covers. Arithmetic and geometric sequences and series, sigma notation, recurrence relations and limits, and the binomial expansion — both for a positive integer index and for negative or fractional indices, where a validity condition applies.

1. Sequences and notation

A **sequence** is an ordered list of terms; a **series** is what you get when you add them. The n th term is written u_n .

Sigma notation

$$\sum_{k=1}^n u_k = u_1 + u_2 + \cdots + u_n$$

The letter below is the starting index, the number above is the last. Two standard results:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n (2k-1) = n^2$$

Worked example. Write $3 + 6 + 9 + \cdots + 30$ in sigma notation.

$$\sum_{k=1}^{10} 3k$$

Recurrence relations

A **recurrence relation** defines each term from the previous one, together with a starting value — for example $u_{n+1} = u_n + 3$ with $u_1 = 2$.

If the sequence converges to a limit L , then replacing both u_{n+1} and u_n by L gives an equation for L .

Worked example. Find the limit of $u_{n+1} = \sqrt{2 + u_n}$.

$$L = \sqrt{2 + L} \implies L^2 = 2 + L \implies L^2 - L - 2 = 0 \implies (L - 2)(L + 1) = 0$$

The terms are positive, so $L = 2$ (rejecting $L = -1$).

Always check the rejected root against the context. A limit of a sequence of square roots cannot be negative, and saying so explicitly is worth a mark.

2. Arithmetic sequences and series

An **arithmetic** sequence has a constant **common difference** d between consecutive terms.

$$u_n = a + (n - 1)d, \quad S_n = \frac{n}{2}(2a + (n - 1)d) = \frac{n}{2}(a + l)$$

where a is the first term and l the last.

Worked example. Find the sum of the first 20 terms with $a = 3$, $d = 5$.

$$S_{20} = \frac{20}{2}(2(3) + 19(5)) = 10(6 + 95) = 1010$$

Worked example. How many terms are in $5, 9, 13, \dots, 101$?

$$a + (n - 1)d = 101 \implies 5 + 4(n - 1) = 101 \implies n - 1 = 24 \implies n = 25$$

Worked example. An arithmetic sequence has 5th term 17 and 10th term 32. Find a and d .

The gap between the 5th and 10th terms is $5d = 32 - 17 = 15$, so $d = 3$. Then $a = 17 - 4(3) = 5$.

Worked example. An arithmetic series has $a = 20$, $d = -3$. For which n is S_n greatest?

The sum grows while the terms are positive. $u_n = 20 - 3(n - 1) > 0$ gives $23 > 3n$, so $n < 7.67$. At $n = 7$ the term is $20 - 18 = 2$ (still positive); at $n = 8$ it is -1 . So S_n is greatest at $n = 7$.

" n th term" and "sum to n terms" are constantly mixed up under time pressure. Read the question twice: u_n is a single term, S_n is a running total.

3. Geometric sequences and series

A **geometric** sequence has a constant **common ratio** r : each term is the previous one multiplied by r .

$$u_n = ar^{n-1}, \quad S_n = \frac{a(1 - r^n)}{1 - r} \quad (r \neq 1)$$

Worked example. For $a = 2$, $r = 3$, find S_5 .

$$S_5 = \frac{2(1 - 3^5)}{1 - 3} = \frac{2(1 - 243)}{-2} = 242$$

Sum to infinity

If $|r| < 1$ the series **converges** and

$$S_\infty = \frac{a}{1 - r}$$

If $|r| \geq 1$ it **diverges** and no sum to infinity exists.

Worked example. Find the sum to infinity of $8 + 4 + 2 + \dots$

Here $a = 8$ and $r = \frac{1}{2}$, and $|r| < 1$, so

$$S_{\infty} = \frac{8}{1 - \frac{1}{2}} = 16$$

Worked example. A geometric series has $S_{\infty} = 3a$. Find r .

$$\frac{a}{1 - r} = 3a \implies \frac{1}{1 - r} = 3 \implies 1 - r = \frac{1}{3} \implies r = \frac{2}{3}$$

State the condition $|r| < 1$ whenever you use S_{∞} . MEI mark schemes award a mark for it, and using the formula on a divergent series produces a plausible-looking but meaningless number.

4. Binomial expansion: positive integer index

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

Equivalently, using binomial coefficients,

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$$

For a positive integer n the expansion **terminates**, giving exactly $n + 1$ terms. The coefficients are the rows of Pascal's triangle; $\binom{n}{0} = \binom{n}{n} = 1$.

Worked example. Find the coefficient of x^3 in $(1 + 2x)^5$.

The required term is $\binom{5}{3}(2x)^3 = 10 \times 8x^3 = 80x^3$, so the coefficient is 80.

Worked example. Find the coefficient of x^2 in $(2 + x)^6$.

$$\binom{6}{2} 2^4 x^2 = 15 \times 16x^2 = 240x^2$$

When the bracket is $(1 + 2x)$ or $(2 + x)$, the whole term must be raised to the power — $(2x)^3 = 8x^3$, not $2x^3$. Forgetting to cube the 2 is the standard error.

Worked example. Find the coefficient of x^3 in $(1 + x)(1 + 2x)^4$.

First expand $(1 + 2x)^4 = 1 + 8x + 24x^2 + 32x^3 + 16x^4$.

Terms in x^3 come from $1 \times 32x^3$ and $x \times 24x^2$:

$$32 + 24 = 56$$

5. Binomial expansion: negative and fractional index

For any real n , the same series formula holds, but the expansion is **infinite** and valid only when $|x| < 1$:

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

Worked example. Expand $(1 + x)^{1/2}$ as far as the term in x^2 .

$$1 + \frac{1}{2}x + \frac{\frac{1}{2}(-\frac{1}{2})}{2}x^2 = 1 + \frac{x}{2} - \frac{x^2}{8}$$

Worked example. Expand $(1 + 3x)^{-2}$ as far as the term in x^2 .

$$1 + (-2)(3x) + \frac{(-2)(-3)}{2}(3x)^2 = 1 - 6x + 3(9x^2) = 1 - 6x + 27x^2$$

The validity condition

The expansion of $(1 + ax)^n$ for non-integer n is valid when $|ax| < 1$, that is $|x| < \frac{1}{|a|}$.

Worked example. State the validity of the expansion of $(1 + 2x)^{1/2}$.

Valid for $|2x| < 1$, i.e. $|x| < \frac{1}{2}$.

The validity condition is worth its own mark and is very often omitted. It is $|x| < \frac{1}{|a|}$, **not** $|x| < 1$, whenever the bracket contains a coefficient on x .

Approximations

Worked example. Use the expansion of $(1 + x)^{1/2}$ to approximate $\sqrt{1.02}$.

Take $x = 0.02$:

$$\sqrt{1.02} \approx 1 + \frac{0.02}{2} - \frac{(0.02)^2}{8} = 1 + 0.01 - 0.00005 = 1.00995$$

For a bracket not starting with 1, factorise first: $\sqrt{4 + x} = 2 \left(1 + \frac{x}{4}\right)^{1/2}$. Applying the series directly to $(4 + x)^{1/2}$ is invalid.

6. Common pitfalls

- Confusing u_n with S_n .
- Using S_∞ without checking, or without stating, that $|r| < 1$.
- Forgetting to raise the coefficient as well as x : $(2x)^3 = 8x^3$.

- Omitting the validity condition for a non-integer index, or giving $|x| < 1$ when it should be $|x| < \frac{1}{|a|}$.
- Applying the infinite series to a bracket not of the form $(1 + \dots)$ without factorising first.
- Off-by-one errors in the number of terms: $u_n = a + (n - 1)d$, not $a + nd$.
- Missing one of the contributing terms when a binomial is multiplied by another bracket.